

Context-Aware Conversational Layout Recommendation for Personalized Interior Design

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I. INTRODUCTION

Interior design and furniture arrangement require balancing user preferences, spatial constraints, and design principles to create functional and aesthetically pleasing spaces. However, many users lack formal design expertise and often struggle to translate their preferences into effective room layouts. Recent advances in artificial intelligence have enabled automated interior layout generation through rule-based optimization and generative models. Despite these developments, most existing systems operate as one-shot layout generators, assuming that users can fully specify their requirements through a single prompt.

In practice, interior design is often an iterative and exploratory process, where users refine their preferences while interacting with possible layout options. Conversational Recommender Systems (CRSs) provide a natural solution to this problem by engaging users in multi-turn dialogue to dynamically elicit preferences and refine recommendations. While CRSs have been widely applied in domains such as e-commerce and entertainment, their application to spatial design tasks such as furniture layout recommendation remains largely unexplored. To address this gap, we propose a Context-Aware Conversational Layout Recommendation framework that integrates spatial room modeling, conversational preference elicitation, and iterative layout refinement. The framework operates in three phases.

Phase 1: The system constructs a structured room representation from user-provided images, dimensions, and constraints while collecting high-level design preferences.

Phase 2: A confidence-driven conversational recommender system interacts with the user to refine preferences and recommend suitable layouts from a generated candidate set.

Phase 3: The system supports iterative refinement, allowing users to request layout modifications that are regenerated and re-evaluated through the conversational loop until a satisfactory design is achieved.

The main contributions of this work are as follows. First, we formulate interior layout design as a conversational recommendation problem. Second, we propose a three-phase framework combining room modeling, conversational reasoning, and layout generation. Finally, Our system focuses on confidence-driven dialog and iterative refinement to progressively converge to user-preferred layouts.

II. RELATED WORKS

In recent years, Conversational Recommender Systems (CRSs) have surged in popularity by transforming traditional, static recommendation engines into dynamic, interactive experiences. Unlike static models that rely purely on historical data and struggle to understand the exact reasons behind a user's preferences, CRSs solve this information asymmetry by eliciting dynamic user needs through real-time, multi-turn natural language interactions. As highlighted in recent comprehensive surveys [3], the development of CRSs involves complex challenges across multiple disciplines, including natural language understanding, multi-turn dialogue strategies, and proactive question-based preference elicitation.

While CRSs have been extensively explored and successfully deployed in domains such as e-commerce, music, and movie recommendations, their application to complex, spatially aware tasks remains largely untouched. Specifically, furniture placement and interior design represent a distinct gap in the current literature. Most existing spatial layout systems lack the proactive, conversational intelligence required to guide users through the nuanced, multi-turn process of interior design. As demonstrated in the following subsections, recent works in furniture layout tend to operate either as one-shot synthesizers or passive executors, highlighting the need for a true CRS that can dynamically consult, recommend, and explain spatial design choices to users.

A. *INTERIORAGENT: LLM Agent for Interior Design-Aware 3D Layout Generation*

In the **INTERIORAGENT** framework [4], 3D furniture layout generation is framed as a code-synthesis problem. Given a natural language prompt, an LLM-based program synthesizer generates an executable scene program using a custom Interior Design-aware Scene Description Language (IDSDL) to place furniture within a room. The resulting scene is then evaluated against a two-tiered set of interior design rules: hard physical constraints are checked mathematically (e.g., *ClearanceConstraint* for walkway space, *CollisionConstraint* to prevent overlapping furniture), while aesthetic and relational rules are assessed by a Vision-Language Model (VLM) (e.g., *ConversationConstraint* to ensure seating faces each other). This multimodal feedback is fed back into the agent to iteratively refine the layout.

A key distinction from our work lies in the interaction paradigm. INTERIORAGENT assumes the user already has a clear design vision and can fully articulate their requirements in a single initial prompt, with no support for conversational refinement or proactive recommendations. Our approach, by contrast, operates as a Conversational Recommender System (CRS) that engages users in multi-turn dialogue to elicit underlying preferences, dynamically recommends furniture placements, and explains the design rationale behind each suggestion, thereby assisting users who may lack formal design knowledge.

B. Chat2Layout: Interactive 3D Furniture Layout with a Multimodal LLM

Chat2Layout [9] introduces an interactive framework powered by a Multimodal LLM (MLLM) for 3D spatial arrangement. The system operates through a continuous perception–reasoning–action loop: it receives user requirements alongside text descriptions and visual captures of the current scene, then employs Chain-of-Thought prompting to decompose high-level requests into atomic operations like adding, removing, translating, scaling, and rotating furniture. Decision-making is further supported by a “Vision-Question Paradigm” combining visual-text prompting with an Offline-to-Online Search mechanism that retrieves reference layouts from a support set. The resulting actions update the scene, enabling iterative user feedback.

While Chat2Layout establishes a genuine multi-turn conversational interface, it functions primarily as an obedient executor rather than a proactive design consultant, still relying on the user to dictate each layout decision step by step. Our work addresses this gap: rather than following user instructions passively, our model analyzes the space and ongoing dialogue to proactively recommend optimal furniture placements, offering design-aware suggestions with explicit rationale and guiding users who may lack a clear initial vision or formal design expertise.

C. DecoMind: A Generative AI System for Personalized Interior Design Layouts

DecoMind [1] introduces a generative AI framework for producing personalized interior design layouts based on user-provided inputs such as room type, design style, and furniture preferences. The system first employs CLIP to retrieve relevant furniture items from a dataset that match the user’s textual description. A preliminary layout containing these furniture items is then combined with a prompt and fed into a Stable Diffusion model with ControlNet to generate a visually coherent interior design that integrates the selected furniture. The generated designs are subsequently evaluated using classifiers to ensure that the final output aligns with the user’s specified preferences.

However, the system primarily focuses on visual scene generation rather than recommendation or interactive design assistance. It assumes that the user can provide all necessary

preferences upfront and lacks mechanisms for multi-turn preference elicitation or conversational refinement. Furthermore, the approach emphasizes image synthesis rather than explicit spatial reasoning or constraint-aware layout generation. In contrast, our work formulates interior layout design as a CRS, enabling iterative dialogue with users to dynamically discover preferences while recommending context-aware furniture placements.

D. Interactive Context-Aware Furniture Recommendation using Mixed Reality

Interactive Context-Aware Furniture Recommendation using Mixed Reality [11] presents a system that leverages Mixed Reality (MR) devices such as HoloLens to recommend and visualize furniture within a user’s physical environment. The system captures spatial context from the surrounding scene and allows users to specify furniture categories through voice commands or gestures. Recommended furniture items are then placed virtually within the environment, enabling users to interact with them directly such as moving or resizing objects to better visualize how they fit within the space.

While the framework effectively integrates spatial context with immersive visualization, it primarily operates as a visual recommendation and interaction tool rather than a conversational design assistant. The system focuses mainly on furniture category selection and placement visualization, without incorporating deeper contextual factors such as user preferences, room attributes, or cultural design constraints. In contrast, our approach employs a CRS that dynamically gathers user preferences and generates layout recommendations while considering spatial constraints and design guidelines.

E. Advancing User-Centric Evaluation and Design of Conversational Recommender Systems

Advancing User-Centric Evaluation and Design of CRSs [6] authors focus on improving the design and evaluation of CRSs from a holistic user-centered perspective. They focus on user-centric evaluation of CRS, emphasizing factors such as empathy, trust, and interaction quality rather than purely system-level metrics.

However, they focus on general CRS evaluation rather than domain-specific tasks involving spatial reasoning and layout generation.

F. LayoutGPT: Compositional Visual Planning and Generation with Large Language Models

To address the challenge of burden on users to provide intricate constraints for visual generation, **LayoutGPT** [2] proposes a collaborative framework that uses LLMs as intermediate visual planners. The methodology uses a sequential pipeline where input text prompt is first processed by an LLM using CSS-based in-context learning. This helps the LLM translate abstract relations into structured layout arrangements. The model predicts unmentioned contexts and generates fine-grained descriptions to enrich the scene. These descriptions act

as spatial constraints for 2D images and 3D scene generative models.

Framing spatial planning as a language modeling task allows the model to translate high-level text into concrete visual plans. Therefore, LayoutGPT outperforms standard text-to-image baselines in accurately rendering the layout, matching the spatial accuracy of supervised methods.

G. Make it Home: Automatic Optimization of Furniture Arrangement

To automate synthesizing indoor scenes, Yu et al. [12] introduced a framework that synthesizes realistic furniture arrangements using ergonomic factors as constraints. The pipeline extracts spatial relations from baseline layouts, encoding features like visibility and accessibility in them and integrating them into a global cost function optimized via Simulated Annealing. Although this approach handles heavy constraints and flexible conditions, the current design strictly assumes that the room has enough perimeter for wall-adjacent objects. Without sufficient wall space, optimization can settle for a local minimum, leaving wall-bound objects awkwardly detached. To address these limitations, unsupervised learning can be used to make it more robust to spatial relations, and account for properties like interior lighting, style and Vastu.

H. ATISS: Autoregressive Transformers for Indoor Scene Synthesis

ATISS [8] presents a novel autoregressive transformer architecture for synthesizing diverse and plausible indoor furniture layouts, given only the room type and its floor plan. Unlike prior approaches that frame scene synthesis as a sequence generation problem, ATISS treats a room as an *unordered set* of objects, leveraging the permutation equivariance of transformers to remain invariant to object ordering. The model is trained end-to-end using only labeled 3D bounding boxes as supervision and supports interactive use cases such as scene completion, partial room re-arrangement, and object suggestions for partially furnished rooms.

While ATISS demonstrates strong performance in generating realistic and diverse layouts, outperforming existing methods while being up to $8\times$ faster, it operates as a purely generative model without any conversational or interactive preference-elicitation mechanism. Users cannot iteratively refine layouts through dialogue, nor does the system proactively recommend placements based on user-expressed preferences. In contrast, our proposed framework embeds such a layout generation capability within a Conversational Recommender System (CRS), enabling multi-turn dialogue to dynamically elicit user preferences and guide spatial design decisions.

I. Interactive Interior Design Recommendation via Coarse-to-fine Multimodal Reinforcement Learning

IIDRS [13] proposes an Interactive Interior Design Recommendation System that mines homeowners' hidden preferences through iterative user interaction using Reinforcement Learning (RL). Rather than relying on a single upfront specification,

the system engages users in multiple rounds of interaction where users provide textual feedback on the gap between the recommended decoration plan and their ideal one. To handle the large action space of possible furniture decorations efficiently, the authors introduce a Decoration Recommendation Coarse-to-Fine Policy Network (DecorRCFN), a two-stage policy structure consisting of a coarse-grained module that narrows down candidate actions and a fine-grained module that models multimodal state-action interactions to make the final selection. Additionally, an object-aware feedback generation method is proposed to augment training with diverse and dynamic textual feedback, improving generalization to unseen online scenarios.

While IIDRS introduces a genuinely interactive, feedback-driven recommendation loop and demonstrates strong performance over traditional baselines, it focuses primarily on decoration item selection rather than full spatial layout generation with explicit furniture placement and room constraint reasoning. Furthermore, the system relies on RL-based policy learning rather than conversational dialogue, limiting its ability to proactively guide users or explain design rationale in natural language. In contrast, our proposed framework adopts a Conversational Recommender System (CRS) approach that combines multi-turn natural language dialogue with spatial reasoning to recommend and justify complete furniture arrangements within a room.

J. Other Works

Merrell et al. [5] proposes an interactive furniture layout system that assists users by encoding interior design guidelines into a density function. By evaluating the spatial configurations, their approach improves the quality of arrangements produced by individuals without prior design training. To address the disconnect between scene understanding and individual object reconstruction, Nie et al. [7] proposes an end-to-end framework that jointly recovers room layout, 3D bounding boxes, and object meshes from a single image. By using contextual dependencies between these components, this approach shows that learning these subtasks together improves performance for indoor layout estimation, object detection and mesh reconstruction. To address the issue of integrating recommendation and conversation modules in CRS, Wang et al. [10] proposes UniCRS, a unified framework driven by knowledge-enhanced prompt learning. It uses a fixed pre-trained model, utilizing customized prompts that have knowledge representations, and generated response templates to combine the 2 subtasks. This approach aligns the recommendation and conversation objectives, showing significant improvement in performance compared to prior methods.

III. PROPOSED METHODOLOGY

Fig. 1 illustrates the end-to-end pipeline of our proposed Context-Aware Conversational Layout Recommendation framework, organized into three distinct phases: room construction and layout generation (Phase 1), conversational recommendation (Phase 2), and iterative refinement (Phase 3).

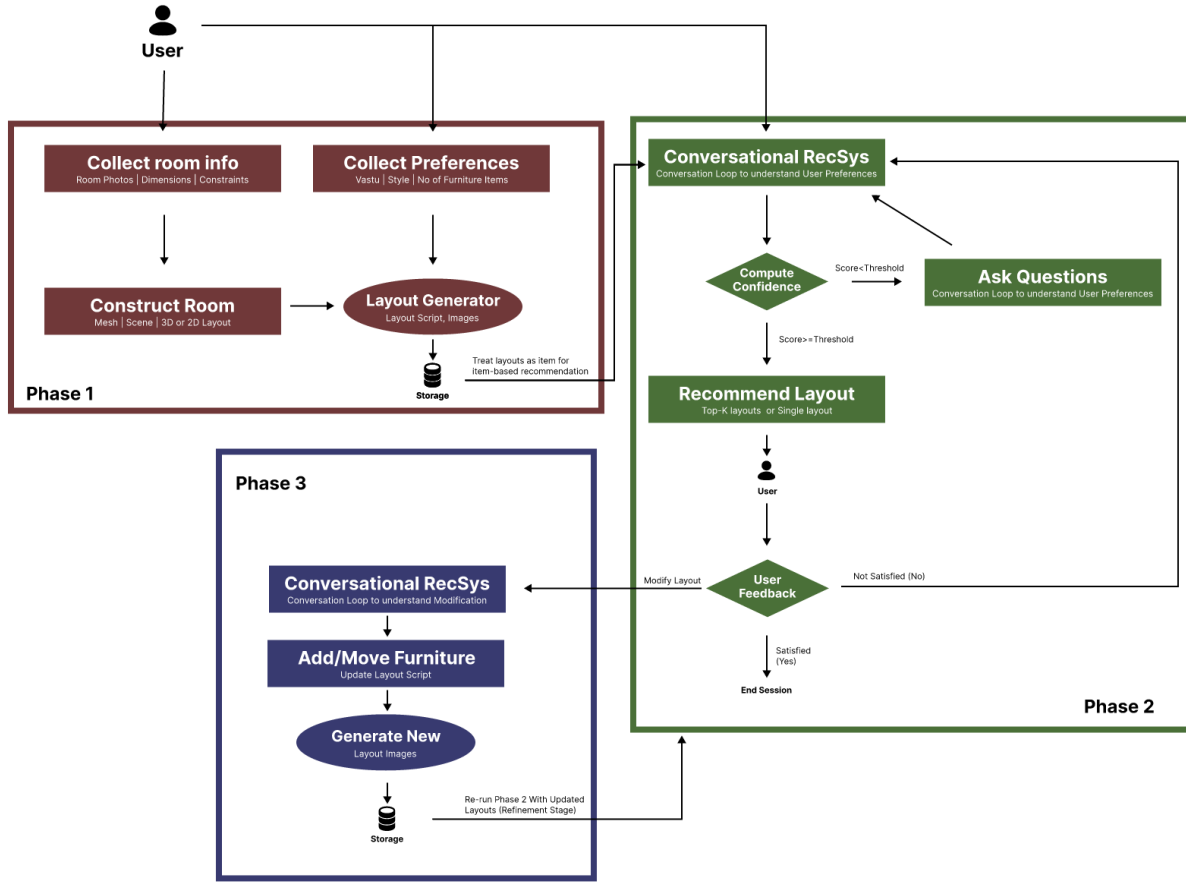


Fig. 1. Overview of the proposed three-phase framework. Phase 1 constructs the room model and generates initial layouts. Phase 2 runs the confidence-driven conversational recommendation loop. Phase 3 handles user-directed iterative refinement, feeding updated layouts back into Phase 2.

A. Phase 1: Room Construction and Layout Generation

Phase 1 is responsible for building a structured spatial representation of the target room and generating an initial set of candidate layouts. It consists of two parallel input streams that converge at the layout generation stage.

The first stream handles *room sensing*. The user provides room photographs, physical dimensions, and spatial constraints. These inputs are processed to construct a digital room model in the form of a mesh, a 3D or 2D scene representation, and a scene graph encoding the geometric properties of the space.

The second stream handles *preference collection*. The user specifies high-level design preferences, including Vastu guidelines, desired style, and the number and type of furniture items to be placed.

Both streams feed into the *Layout Generator*, which produces a set of candidate layout scripts and layout images. These layouts are stored and subsequently treated as items for recommendation in Phase 2.

B. Phase 2: Conversational Recommendation

Phase 2 forms the core recommendation loop of the system, driven by a *Conversational Recommender System (CRS)* that dynamically elicits and refines user preferences through multi-turn dialogue.

Upon receiving the generated layouts and user context from Phase 1, the CRS enters a confidence-based decision loop. A confidence score is computed over the current understanding of the user's preferences. If the score falls below a predefined threshold, the system invokes an *Ask Questions* module, which conducts a targeted conversational exchange to gather additional preference information. This loop continues until the confidence score exceeds the threshold, at which point the system proceeds to recommendation.

The *Recommend Layout* module then surfaces the top- K candidate layouts, or a single best layout, to the user. The user reviews the presented options and provides explicit feedback via a *User Feedback* decision gateway. If the user is satisfied, the session terminates, and the chosen layout is returned as the

final output. If the user is not satisfied, the system transitions to Phase 3 for iterative refinement.

C. Phase 3: Iterative Refinement

Phase 3 handles the case where the user wishes to modify the recommended layout. A dedicated *Conversational Recommendation System* module engages the user in a focused dialogue to understand the nature of the desired modifications, such as repositioning furniture, swapping items, or adjusting spatial arrangements.

Based on the gathered modification intent, the *Add/Move Furniture* module updates the layout script accordingly. A *Generate New* step then produces fresh layout images reflecting the requested changes, which are stored and fed back into Phase 2 as an updated item set. Phase 2 is then re-executed with the refined layouts, forming a closed refinement loop that iterates until the user is satisfied.

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